
Predicting Colour Yield from Process Control Limits

A process capability study of anodizing and
PVD thin film surface finishing

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Summary of purpose. This report examines how tightly a surface finishing line must hold its process settings in order to produce parts of a consistent colour. Two industrial routes are modelled from first principles: anodizing with dyeing, and physical vapour deposition of thin films. The study quantifies, for each route, what fraction of parts fall inside a colour specification and which single process control most deserves investment.

Key findings at a glance

Yield of the PVD interference stack at realistic tolerances	56.4 %
Colour capability index of that stack, C_{pk}	−0.018
Share of its colour variance caused by nitrogen stoichiometry alone	69.8 %
Tightening of nitrogen control needed to reach $C_{pk} = 1.33$	6.0×
Tightening of coating thickness control needed	none
Anodizing yield with a black dye	100.0 %
Anodizing yield with a blue dye, process held identical	77.7 %

Executive Summary

Background

Products that must look identical, such as laptop lids, watch cases and phone frames, obtain their colour from a surface finishing process. Every such process drifts during normal operation. An acid bath warms up over a shift, a dye becomes depleted, a gas flow controller reads slightly high. None of these is a fault. The question is whether the resulting colour still falls inside tolerance.

What was done

A physics based computer model was built covering two finishing routes, and realistic process variation was pushed through it to predict colour yield. The model runs the complete causal chain from a change in process setting, through the film that grows, to the light that film reflects, to the colour a human observer perceives, and finally to the fraction of parts that pass inspection.

Principal findings

1. **The PVD interference stack is not capable at realistic tolerances.** It yields 56.4 % against a colour specification of $\Delta E_{00} \leq 1.0$, with a capability index of -0.018 . A negative capability index means the average part is already a reject, not merely that the spread is too wide.
2. **A single variable dominates, and it is not the obvious one.** Nitrogen stoichiometry accounts for 69.8 % of all colour variance. Coating thickness, which is the variable an engineer would instinctively control, accounts for only 2.5 % and never requires tightening. The film becomes optically opaque beyond roughly 150 nm, so additional thickness produces no visible change.
3. **The corrective action is quantified.** Achieving a capability index of 1.33 requires nitrogen control to be tightened six fold, from ± 0.040 to ± 0.0067 , and overcoat thickness control to be tightened 2.7 fold, from ± 8.0 nm to ± 3.0 nm.
4. **The choice of colour is itself a capability decision.** Holding the anodizing process, its tolerances and the sample size completely fixed, and changing only which dye is in the tank, moves yield from 100 % for black to 77.7 % for blue. This 22 percentage point swing originates in a design review, not on the production line.
5. **The two routes fail in opposite places.** Anodized black is robust in colour but limited by film thickness. The PVD stack has no thickness problem at all and is limited entirely by composition. A control plan written for one route is of little use for the other.

Recommendations

1. Assess cosmetic capability at the point of colour selection, before a process is specified, because the colour chosen can dominate every subsequent process improvement.
2. For the PVD route, direct investment towards mass flow control precision and in chamber optical monitoring rather than towards deposition timing.
3. For the anodizing route, direct investment towards bath temperature control and coating thickness metrology.
4. Validate this model experimentally before using it for absolute colour prediction. Its trends are trustworthy; its absolute colour values are not yet confirmed against physical samples.

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1 Purpose and Scope

1.1 The problem this report addresses

Consider any anodized aluminium laptop or any coated stainless steel watch. It has a colour. Now consider that the factory produced a million of them, and every one must be the same colour, close enough that a customer holding two side by side cannot tell them apart.

This is harder than it appears, because the colour is not painted on. It is a by product of a chemical or physical process running inside a tank or a vacuum chamber, and every such process drifts during normal operation. These drifts are not faults; they are the ordinary behaviour of a production line. The engineering question is whether the colour they produce remains inside the tolerance band.

The central question. Given how tightly a real production line holds its process settings, what fraction of parts emerge the correct colour, and if that fraction is insufficient, which single control should be tightened first?

The commercial importance is specific. A part that fails a dimensional check fails at the machine that made it, early, when little value has been added. A part that fails a *cosmetic* check fails at the very end of the line, after all the value has been added, and in most cases it cannot be reworked. Cosmetic rejects are therefore disproportionately expensive.

1.2 Scope

1.2.1 Included in this study

- Type II sulfuric acid anodizing of aluminium, followed by dyeing.
- Physical vapour deposition of decorative nitride thin film stacks.
- Prediction of the reflectance spectrum produced by each route.
- Conversion of that spectrum to perceived colour using the international standard colourimetric system.
- Propagation of process variation through to capability indices and predicted yield.
- Identification and ranking of the process variables that drive colour variation.
- Calculation of the control tolerances required to meet a stated capability target.

1.2.2 Excluded from this study

- Experimental validation. No physical coupons were produced or measured.
- Mechanical properties such as hardness, adhesion and wear resistance.
- Corrosion performance and accelerated ageing.
- Gloss, texture and surface roughness, which contribute to appearance but are treated here as held constant.
- Cost modelling and equipment selection.
- Long term process drift between batches, shifts and operators. Only within batch variation is modelled. Section 7.3 explains why this matters.

1.3 How to read this report

Section 2 explains the necessary background from first principles and assumes no prior knowledge of surface finishing, optics or statistics. Readers already familiar with the field may proceed directly to Section 3, which describes the model. Every equation in Section 3 is followed by a complete definition of its symbols, their units and their typical magnitudes. Section 4 records how the implementation was verified. Sections 5 and 6 present and discuss the results. Section 7 states the limitations precisely, and Section 8 gives conclusions and recommendations. A complete list of symbols appears in Appendix A.

2 Background

This chapter builds up the four ideas the report depends upon: how anodizing works, how thin film coating works, how colour is measured numerically, and how process capability is measured. Each is presented without assuming prior exposure.

2.1 Anodizing

Anodizing is an electrochemical process. An aluminium part is immersed in an acid bath, usually dilute sulfuric acid, and connected as the positive electrode. Passing current causes the aluminium surface to oxidise, growing

a hard, porous oxide skin out of the metal's own surface rather than depositing a separate layer on top of it.

Two features of this oxide matter for colour:

1. **It is porous.** The film contains a dense array of fine vertical pores. These pores can be filled with dye, which is how anodized parts obtain colour. More pore volume means more dye can be absorbed.
2. **It is transparent.** Aluminium oxide does not absorb visible light. Light therefore passes through the film, reflects from the metal beneath, and passes back out. The colour is produced entirely by whatever dye sits in the pores.

Type II refers to conventional sulfuric acid anodizing producing films of roughly 5 to 25 μm thickness, which is the standard architectural and decorative process. Type I uses chromic acid and Type III, often called hard anodizing, uses colder baths and higher voltages to produce thicker, denser and more wear resistant films.

2.2 Physical vapour deposition

Physical vapour deposition, abbreviated PVD, is a vacuum process. A solid metal target, commonly titanium, is vaporised by bombardment. The resulting vapour travels across the chamber and condenses onto the part as an extremely thin film, typically a few hundred nanometres thick, which is roughly one thousandth the width of a human hair.

If nitrogen gas is admitted during deposition, the metal vapour reacts with it to form a nitride. Titanium nitride is a hard ceramic which happens to be gold coloured, which is why it is widely used on watch cases, taps and cutting tools.

The colour of a PVD coating arises from two distinct mechanisms, and distinguishing them is essential to understanding the results in Section 5:

1. **Bulk absorption.** The nitride itself absorbs some wavelengths more than others. Titanium nitride absorbs blue light strongly and reflects red light well, which is what makes it look gold. This mechanism depends on the material's *composition*, not on how much of it is present, provided there is enough to be opaque.
2. **Optical interference.** Because the layers are comparable in thickness to the wavelength of visible light, light reflected from the top surface of a layer can interfere with light reflected from its bottom surface. The two waves reinforce at some wavelengths and cancel at others. This mechanism depends strongly on layer *thickness*, and it is the same effect that produces the colours in a soap bubble or an oil film on water.

2.3 Measuring colour numerically

2.3.1 The reflectance spectrum

The most complete physical description of how a surface interacts with light is its **reflectance spectrum**, written $R(\lambda)$. This states, for each wavelength λ of light, what fraction is reflected. A value of $R = 0.8$ at 600 nm means the surface reflects 80 % of the orange light striking it.

An object appears red because its spectrum is high in the red region of the spectrum, around 650 nm, and low elsewhere. Visible light spans roughly 380 nm, which appears violet, to 780 nm, which appears deep red.

2.3.2 From spectrum to perceived colour

A spectrum is not itself a colour. Colour is a perception, and the human eye does not record the full spectrum. It contains three types of cone cell, each sensitive to a broad and overlapping band of wavelengths. The eye therefore compresses the entire spectrum into just three numbers.

The Commission Internationale de l'Éclairage, abbreviated CIE, standardised this in 1931 by defining three **colour matching functions**, written $\bar{x}(\lambda)$, $\bar{y}(\lambda)$ and $\bar{z}(\lambda)$. These describe the sensitivity of a standard human observer at each wavelength. Weighting a spectrum by these functions and summing produces three numbers called tristimulus values, from which every further colour calculation follows.

2.3.3 CIELAB

Tristimulus values are not convenient for engineering because equal numerical differences do not correspond to equal visible differences. The CIELAB system addresses this by transforming them into three more useful coordinates:

L^*	Lightness , from 0 for perfect black to 100 for perfect white.
a^*	Green to red axis . Negative values are green, positive values are red, zero is neutral.
b^*	Blue to yellow axis . Negative values are blue, positive values are yellow, zero is neutral.

A saturated gold finish might be described as $L^* = 76$, $a^* = 4$, $b^* = 49$, meaning fairly light, very slightly red, and strongly yellow.

2.3.4 Colour difference

To state whether two parts match, a single number is needed. The current international standard is **CIEDE2000**, written ΔE_{00} . Its interpretation is approximately as follows:

$\Delta E_{00} < 0.5$	Not distinguishable even under careful side by side comparison.
$\Delta E_{00} \approx 1.0$	The threshold at which a trained observer can reliably detect a difference. Most consumer product cosmetic specifications are set here or tighter.
$\Delta E_{00} \approx 2$ to 3	Visible to an ordinary observer comparing two parts.
$\Delta E_{00} > 5$	Obviously different colours.

This report uses $\Delta E_{00} \leq 1.0$ as the colour specification throughout.

2.4 Measuring process capability

2.4.1 Why an average is not enough

Knowing that a process produces the correct colour *on average* says nothing useful. What matters is the spread. A process centred perfectly in the middle of its tolerance but with a very wide spread will produce many rejects, while a process slightly off centre but with a very tight spread may produce none.

2.4.2 The capability index

The capability index C_{pk} combines centring and spread into one number. It compares the distance from the process mean to the nearest specification limit against three standard deviations of the process spread.

$C_{pk} < 1.0$	Not capable. The process spread is wider than the tolerance band, and rejects are certain.
$C_{pk} = 1.0$	Marginal. The tolerance band is exactly as wide as the process spread, giving roughly 2700 defects per million.
$C_{pk} = 1.33$	The usual consumer electronics gate, giving roughly 32 defects per million. This report adopts it as the target.
$C_{pk} = 2.0$	Very comfortable, roughly 2 defects per billion.
$C_{pk} < 0$	The process mean lies outside the specification limit entirely. The average part is a reject.

2.4.3 Design of experiments

Investigating several process settings by varying one at a time is inefficient and, more seriously, cannot detect **interactions**, meaning cases where the effect of one variable depends on the level of another. A design of experiments, abbreviated DOE, is a structured plan that varies several settings simultaneously according to a pattern chosen so that individual effects, curvature and interactions can all be separated from a modest number of runs.

3 Modelling Approach

This chapter states the model completely. Every equation is followed by a definition of each symbol it contains, the units of that symbol, and where useful a typical magnitude. A consolidated symbol list appears in Appendix A.

3.1 The shared pipeline

The two finishing routes are modelled independently up to the point at which each produces a reflectance spectrum. From there they share a single common path to colour and capability:

$$\text{process variation} \rightarrow \text{physics} \rightarrow R(\lambda) \rightarrow L^*a^*b^* \rightarrow \Delta E_{00} \rightarrow C_{pk} \rightarrow \text{yield}$$

Sharing the colour engine is deliberate and is what makes the two routes directly comparable. Any difference in the final capability figure is attributable to the physics of the finish itself, and not to any difference in how colour was computed.

3.2 The colour engine

3.2.1 Tristimulus values

The first step converts a reflectance spectrum into three numbers representing the response of the human eye.

$$X = k \int_{380}^{780} R(\lambda) S(\lambda) \bar{x}(\lambda) d\lambda \quad (1)$$

with Y and Z defined identically but using $\bar{y}(\lambda)$ and $\bar{z}(\lambda)$ respectively.

What this equation says. At every wavelength, multiply three things together: how much light the illuminant provides, what fraction the surface reflects, and how sensitive the eye is. Then add up the result across the whole visible range. Doing this three times, once for each cone sensitivity, gives three numbers.

- X First tristimulus value. Dimensionless. Loosely corresponds to a red response.
- Y Second tristimulus value. Dimensionless, scaled to 100 for a perfect white. Corresponds to luminance, meaning perceived brightness.
- Z Third tristimulus value. Dimensionless. Loosely corresponds to a blue response.
- λ Wavelength of light, in nanometres. The integral runs from 380 to 780 nm, the visible range. The computation uses 5 nm steps, which is the CIE recommended interval.
- $R(\lambda)$ Reflectance factor of the surface at wavelength λ . Dimensionless, ranging from 0 for a perfect absorber to 1 for a perfect reflector. This is the output of the physics models in Sections 3.3 and 3.4.
- $S(\lambda)$ Relative spectral power distribution of the illuminant. Dimensionless. This report uses CIE Standard Illuminant D65, which represents average daylight at a correlated colour temperature of approximately 6500 K. The illuminant matters because an object's colour depends on the light falling on it.
- $\bar{x}, \bar{y}, \bar{z}$ CIE 1931 2 degree colour matching functions. Dimensionless. These are tabulated international standards describing the sensitivity of a standard human observer.
- k Normalising constant, defined in Equation (2). It scales the result so that a perfect white reflector returns $Y = 100$.

$$k = \frac{100}{\int_{380}^{780} S(\lambda) \bar{y}(\lambda) d\lambda} \quad (2)$$

Why this normalisation exists. Without it, the tristimulus values would depend on how bright the illuminating lamp happened to be. Dividing through by the response to the illuminant alone removes that dependence, so the numbers describe the *surface* rather than the lighting.

3.2.2 Conversion to CIELAB

Tristimulus values are converted to the perceptually organised CIELAB coordinates by

$$L^* = 116 f(Y/Y_n) - 16 \quad (3)$$

$$a^* = 500 [f(X/X_n) - f(Y/Y_n)] \quad (4)$$

$$b^* = 200 [f(Y/Y_n) - f(Z/Z_n)] \quad (5)$$

where the compressing function f is

$$f(s) = \begin{cases} s^{1/3}, & s > \left(\frac{6}{29}\right)^3 \\ \frac{1}{3} \left(\frac{29}{6}\right)^2 s + \frac{4}{29}, & \text{otherwise} \end{cases} \quad (6)$$

What these equations say. Each tristimulus value is first divided by the corresponding value for a reference white, then passed through a cube root. Lightness comes from the Y channel alone. The two colour axes come from differences between channels, which is why a^* and b^* are both zero for any neutral grey.

L^* Lightness, 0 to 100. Dimensionless.

a^*, b^* Chromatic coordinates. Dimensionless, typically in the range -128 to $+127$.

X_n, Y_n, Z_n Tristimulus values of the reference white, meaning a perfect reflecting diffuser under the same illuminant. For D65 with the 2 degree observer these are 95.047, 100.000 and 108.883.

$f(s)$ Compressing function of Equation (6). The cube root reproduces the eye's non linear response to brightness: doubling the light reaching the eye does not double the perceived lightness. The linear branch avoids the infinite slope the cube root would otherwise have near zero, which would make very dark colours numerically unstable.

3.2.3 Colour difference

Two colours are compared using the CIEDE2000 formula:

$$\Delta E_{00} = \sqrt{\left(\frac{\Delta L'}{k_L S_L}\right)^2 + \left(\frac{\Delta C'}{k_C S_C}\right)^2 + \left(\frac{\Delta H'}{k_H S_H}\right)^2 + R_T \left(\frac{\Delta C'}{k_C S_C}\right) \left(\frac{\Delta H'}{k_H S_H}\right)} \quad (7)$$

What this equation says. At its core it is a distance formula, the three dimensional equivalent of Pythagoras' theorem, measuring how far apart two colours are in lightness, colour intensity and hue. The complication is that the eye is not equally sensitive everywhere in colour space. It detects a small shift in a pale grey far more readily than the same numerical shift in a saturated blue. Each term is therefore divided by a weighting function that accounts for this, and a fourth term corrects an additional distortion in the blue region.

$\Delta L'$ Difference in lightness between the two colours.

$\Delta C'$ Difference in chroma, meaning colour intensity or how far the colour sits from neutral grey.

$\Delta H'$ Difference in hue, meaning position around the colour circle from red through yellow, green and blue.

S_L, S_C, S_H Weighting functions for lightness, chroma and hue. Each increases where the eye is less sensitive, which reduces the contribution of that term. For example $S_C = 1 + 0.045 \bar{C}'$ grows with chroma, encoding the fact that saturated colours tolerate larger shifts before the difference becomes visible.

R_T Rotation term. Corrects an interaction between chroma and hue that occurs specifically in the blue region, where the axes of perceived colour space are tilted relative to the coordinate axes. It is essentially zero outside the blues.

k_L, k_C, k_H Parametric factors allowing adjustment for unusual viewing conditions. All are set to 1 in this report, which is the standard reference condition.

The implementation follows Sharma, Wu and Dalal (2005), which resolves ambiguities in the original CIE specification concerning how hue angles are averaged when they straddle the 0 and 360 degree boundary. These edge cases are the most common source of error in CIEDE2000 implementations, which is why the verification in Section 4 tests them explicitly.

3.3 Modelling the anodizing route

3.3.1 Film growth

Anodic oxide grows electrochemically from the applied current while the acid simultaneously dissolves it back. The net rate is the difference between these two competing processes:

$$\frac{dx}{dt} = \underbrace{\frac{M_{\text{ox}} j \eta}{z F \rho_{\text{ox}}}}_{\text{electrochemical growth}} - \underbrace{k_0 \exp\left(-\frac{E_a}{R_g T}\right) [\text{H}_2\text{SO}_4]^m}_{\text{chemical back dissolution}} \quad (8)$$

What this equation says. The first term is Faraday's law of electrolysis, which states that the amount of material converted is proportional to the electric charge passed. The second term is an Arrhenius rate law describing ordinary chemical attack of the oxide by the acid, which proceeds whether or not current is flowing. The film grows only as fast as the first term exceeds the second.

- x Oxide film thickness, in metres. Reported throughout in micrometres, where $1 \mu\text{m} = 10^{-6} \text{m}$. Typical Type II values are 5 to $25 \mu\text{m}$.
- t Time, in seconds. Reported in minutes, with typical anodizing times of 20 to 60 minutes.
- M_{ox} Molar mass of aluminium oxide, Al_2O_3 , equal to 0.10196 kg/mol .
- j Current density, in amperes per square metre. Reported in A/dm^2 , the industry convention, where $1 \text{ A/dm}^2 = 100 \text{ A/m}^2$. Typical Type II operation is 1.0 to 2.0 A/dm^2 .
- η Current efficiency, dimensionless, between 0 and 1. The fraction of the charge passed that ends up as retained oxide rather than being lost to oxygen evolution or to immediate re-dissolution at the pore base. Typically 0.6 to 0.8, and modelled here as falling with both temperature and acid strength.
- z Number of electrons transferred per formula unit of oxide formed, equal to 6. Forming one Al_2O_3 unit requires oxidising two aluminium atoms, each surrendering three electrons.
- F Faraday constant, $96\,485 \text{ C/mol}$. The electric charge carried by one mole of electrons.
- ρ_{ox} Density of the anodic oxide, taken as 3100 kg/m^3 . This is deliberately below the 3970 kg/m^3 of dense sapphire because the anodic film is porous and partly hydrated.
- k_0 Pre-exponential factor of the dissolution rate law, in metres per second per gram per litre. Calibrated so that a bath at 20°C and 180 g/L dissolves approximately $0.05 \mu\text{m}$ per minute, the accepted figure for Type II baths.
- E_a Activation energy for chemical dissolution, taken as $65\,000 \text{ J/mol}$. This value governs how strongly the dissolution rate responds to temperature.
- R_g Universal gas constant, 8.314 J/(mol K) . Written R_g here to avoid confusion with reflectance R .
- T Absolute bath temperature, in kelvin. Reported in degrees Celsius, with typical Type II operation at 18 to 22°C .
- $[\text{H}_2\text{SO}_4]$ Sulfuric acid concentration, in grams per litre. Typical operation is 150 to 200 g/L .
- m Reaction order with respect to acid concentration, taken as 1, meaning the dissolution rate is directly proportional to acid strength.

Why bath temperature is the dominant lever. The dissolution term contains temperature inside an exponential. Evaluating it shows the rate roughly doubles for every 10°C rise. A bath that drifts warm therefore does two damaging things simultaneously: it thins the film, and it widens the pores. The part emerges lighter and duller. No other single variable produces this combined effect, which is why temperature control is the first investment any anodizing line makes.

3.3.2 Porosity and dye uptake

The dye occupies the pores, so the quantity of dye a film can hold is proportional to the available pore volume, which is porosity multiplied by thickness. Uptake approaches saturation exponentially with immersion time:

$$cx \propto \underbrace{Px}_{\text{pore volume}} \times \underbrace{\left(1 - e^{-t_d/\tau}\right)}_{\text{time saturation}} \times \underbrace{\exp\left[-\frac{1}{2}\left(\frac{\text{pH} - 5.5}{1.4}\right)^2\right]}_{\text{pH dependence}} \times \frac{c_b}{c_{b,0}} \quad (9)$$

- cx Dye concentration multiplied by path length, the quantity that actually appears in the Beer-Lambert absorption law of Equation (10). Dimensionless as used here.
- P Porosity, the volume fraction of the film occupied by pores. Dimensionless, typically 0.10 to 0.15 for Type II. Modelled as increasing with both bath temperature and acid concentration.
- t_d Dye immersion time, in minutes. Typically 10 to 15.
- τ Characteristic uptake time constant, in minutes. Shorter in a hotter dye bath because diffusion into the pores is faster.

- pH Acidity of the dye bath. The Gaussian form centred on 5.5 encodes the experimental observation that acid dye adsorption onto the amphoteric oxide surface peaks near this value and falls away on both sides.
- c_b Dye bath concentration, in grams per litre, with $c_{b,0}$ the nominal value of 3 g/L used for normalisation.

3.3.3 Optical response

A dyed anodic film is a **bulk absorber**, not an interference filter. Light crosses the transparent film, reflects from the aluminium beneath, and crosses back. The Beer-Lambert law therefore applies over a double pass:

$$R(\lambda) = R_{\text{surf}} + (1 - R_{\text{surf}})^2 R_{\text{Al}}(\lambda) e^{-2\varepsilon(\lambda) c x} \quad (10)$$

What this equation says. Some light reflects straight off the film's outer surface without ever entering it, and that portion carries no colour. The remainder enters the film, is attenuated on the way down, reflects from the metal, and is attenuated again on the way back up. The factor of 2 in the exponent is that double pass.

R_{surf} Fresnel reflectance of the air to oxide interface, approximately 0.056 for an oxide refractive index near 1.62. This is the portion of light that never enters the film.

$(1 - R_{\text{surf}})^2$ The fraction transmitted into the film and then back out again, hence the square.

$R_{\text{Al}}(\lambda)$ Reflectance of the aluminium substrate, approximately 0.92 across the visible range.

$\varepsilon(\lambda)$ Molar absorptivity of the dye at wavelength λ . Represented as a sum of Gaussian absorption bands whose positions determine the perceived colour.

$c x$ Dye loading from Equation (9).

A dye absorbs the complement of the colour it appears. A blue dye looks blue because it removes orange and red light from the reflected beam. Its absorption band therefore sits near 600 nm, not in the blue. This is why the band positions in the model appear to be in the wrong place at first glance.

Because Equation (10) contains no interference term, a dyed anodic film barely changes colour with viewing angle. This is the exact opposite of the PVD stack described next, and the contrast turns out to matter.

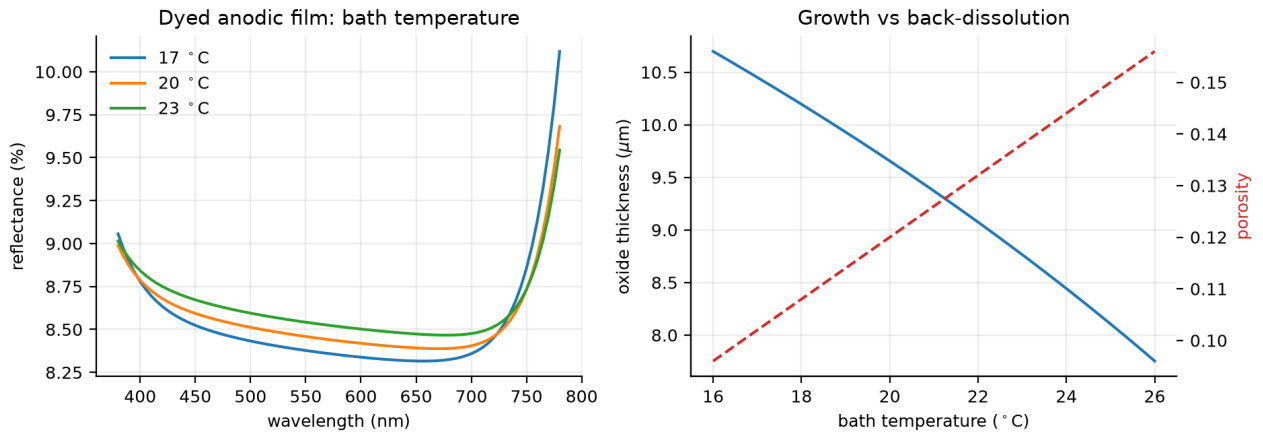


Figure 1: Left: reflectance of a black dyed film across a $\pm 3^\circ\text{C}$ bath excursion. Right: the competition described by Equation (8). As the bath warms, back dissolution accelerates and thickness falls (solid line), while porosity rises (dashed line).

3.4 Modelling the PVD route

3.4.1 The transfer matrix method

Because PVD layers are comparable in thickness to the wavelength of light, the reflection from a stack cannot be computed surface by surface. Multiple reflections within each layer interfere, and all of them must be summed consistently. The standard treatment assigns each layer a characteristic matrix:

$$M_j = \begin{bmatrix} \cos \delta_j & \frac{i \sin \delta_j}{\eta_j} \\ i \eta_j \sin \delta_j & \cos \delta_j \end{bmatrix} \quad (11)$$

What this equation says. The matrix describes how the electric and magnetic fields of a light wave are transformed on passing through one layer. Its usefulness is that the effect of a whole stack is obtained by simply multiplying the individual matrices together, in order.

- M_j Characteristic matrix of layer j , a 2×2 complex matrix.
- δ_j Phase thickness of layer j , defined in Equation (12). It measures how much the wave's phase advances in crossing the layer.
- η_j Optical admittance of layer j , defined in Equation (13). It describes how readily the layer supports the propagating wave.
- i The imaginary unit. Complex arithmetic is essential here because it is what carries the phase information on which interference depends.

The phase thickness and admittance are

$$\delta_j = \frac{2\pi t_j N_j \cos \theta_j}{\lambda} \quad (12)$$

$$\eta_j = \begin{cases} N_j \cos \theta_j, & \text{s polarised} \\ N_j / \cos \theta_j, & \text{p polarised} \end{cases} \quad (13)$$

- t_j Physical thickness of layer j , in nanometres. Typically 80 to 400 nm.
- N_j Complex refractive index of layer j , written $N = n - ik$. See Section 3.4.2.
- θ_j Angle of propagation inside layer j , obtained from Snell's law. It is complex for an absorbing layer, which has no simple geometric interpretation but is mathematically correct.
- λ Wavelength in vacuum, in nanometres.
- s, p The two polarisation states of light, referring to whether the electric field oscillates perpendicular or parallel to the plane of incidence. They reflect differently at non normal incidence, so both are computed and averaged.

The assembled stack matrix maps the substrate admittance to the front surface:

$$\begin{bmatrix} B \\ C \end{bmatrix} = \left(\prod_{j=1}^J M_j \right) \begin{bmatrix} 1 \\ \eta_{\text{sub}} \end{bmatrix}, \quad Y = \frac{C}{B}, \quad R = \left| \frac{\eta_0 - Y}{\eta_0 + Y} \right|^2 \quad (14)$$

- B, C Normalised electric and magnetic field amplitudes at the front surface of the stack. Dimensionless and complex.
- η_{sub} Optical admittance of the substrate.
- η_0 Optical admittance of the surrounding medium, air, for which $n = 1$.
- Y Equivalent admittance of the entire coated surface. This single complex number summarises the whole stack.
- R Reflectance, the quantity fed into Equation (1). Taking the squared magnitude converts the amplitude reflection coefficient into a power fraction, because intensity is proportional to amplitude squared.
- J Number of layers in the stack.

This is the identical calculation performed by the commercial thin film design packages Essential Macleod and TFCALC.

A sign convention that silently destroys the result. Tabulated optical constants are published as $N = n + ik$, which pairs with one convention for how a wave is written mathematically. The matrix formulation above uses the opposite convention and expects $N = n - ik$. Supplying the first to the second converts every absorbing layer into a *gain* medium, one that amplifies light rather than absorbing it. The code still runs, the plots still look plausible, and reflectance quietly exceeds 100 %. This error occurred during development and was caught by an automated check asserting that $0 \leq R \leq 1$ for every stack at every angle. That check is retained permanently, because the failure is otherwise invisible.

3.4.2 Optical constants

Every material is described by a complex refractive index:

$$N(\lambda) = n(\lambda) - i k(\lambda) \quad (15)$$

- n Real refractive index. Describes how much the material slows light, and therefore how much it bends and how strongly it reflects. Air is 1.0, water 1.33, anodic alumina approximately 1.62.
- k Extinction coefficient. Describes how strongly the material absorbs light. Transparent materials such as SiO₂ have k effectively zero. Metals have k of several units, meaning light penetrates only tens of nanometres before being extinguished.

For metals and metallic nitrides, N is obtained from a Drude-Lorentz model of the dielectric function:

$$\epsilon(\omega) = \epsilon_{\infty} - \frac{\omega_p^2}{\omega^2 + i\Gamma\omega} + \sum_j \frac{f_j \omega_j^2}{\omega_j^2 - \omega^2 - i\gamma_j\omega}, \quad N = \sqrt{\epsilon} \quad (16)$$

What this equation says. It splits the material's optical response into two physical contributions. The second term describes free electrons, which are what make a metal shiny and reflective. The third term describes electrons bound in specific energy transitions, each behaving like a damped oscillator with its own resonant frequency. In titanium nitride it is precisely one of these bound transitions, absorbing in the blue, that produces the gold colour.

- ϵ Complex dielectric function. The refractive index is its square root.
- ω Angular frequency of the light, expressed in electronvolts through $E [\text{eV}] = 1239.84/\lambda [\text{nm}]$. The visible range spans roughly 1.6 to 3.3 eV.
- ϵ_{∞} High frequency dielectric constant, accounting for all contributions above the modelled range. Dimensionless, typically 1 to 5.
- ω_p Plasma frequency, in eV. Sets how many free electrons the material has and therefore where its reflectance edge lies. Aluminium has a very high value near 15 eV, which is why it reflects the entire visible spectrum and appears as a neutral mirror.
- Γ Drude damping constant, in eV. Represents how frequently free electrons scatter. Higher values make a metal duller.
- f_j Strength of the j th bound oscillator. Dimensionless.
- ω_j Resonant frequency of the j th oscillator, in eV. This is where that transition absorbs.
- γ_j Damping of the j th oscillator, in eV. Controls how broad the absorption band is.

3.4.3 Stoichiometry as the colour control

Titanium nitride absorbs strongly enough that beyond roughly 150 nm the film is optically opaque and the substrate beneath becomes irrelevant. Thickness therefore stops controlling colour. The variable that does control it is the composition, written x in the formula TiN _{x} , which is set on the deposition tool by the nitrogen partial pressure.

Stoichiometry. $x = 1$ denotes exactly one nitrogen atom per titanium atom, the ideal composition. $x < 1$ denotes a nitrogen starved, metal rich film. $x > 1$ denotes a nitrogen stuffed film.

The model represents the effect of composition through Equation (16) as follows. As x falls below 1, more free carriers are present, so ω_p rises and the reflectance edge moves towards the blue, shifting the appearance from gold towards metallic silver. Simultaneously the bound oscillator responsible for the gold colour weakens, and increased point defect density broadens every spectral feature.

3.5 Propagating process variation

Each process input is assigned a control tolerance representing how tightly the line actually holds it. That tolerance is interpreted as a $\pm 3\sigma$ band, meaning the line holds the stated range essentially all of the time. Values are sampled from a normal distribution and each sample is pushed through the forward physics model. This is a Monte Carlo simulation, using 6000 samples throughout.

Capability is then computed as

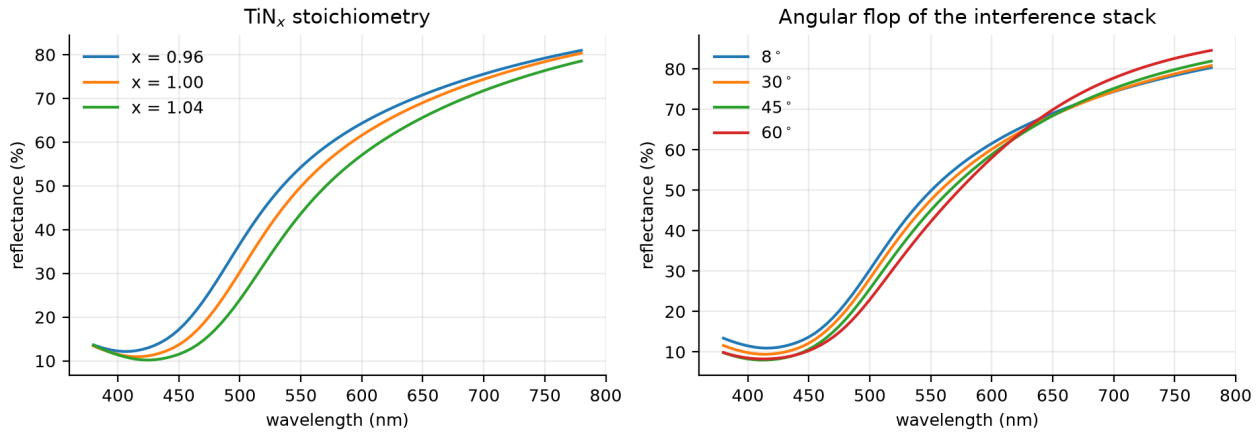


Figure 2: Left: reflectance of the PVD stack across a $\pm 4\%$ swing in nitrogen stoichiometry. Right: angular colour shift of the same interference stack, which is the mechanism by which a part can pass inspection viewed face on and fail when viewed at a chamfer.

$$C_{pk} = \frac{\min(USL - \mu, \mu - LSL)}{3\sigma}, \quad C_{pk}^{\text{upper}} = \frac{USL - \mu}{3\sigma} \quad (17)$$

What this equation says. Measure the distance from the process mean to the nearer specification limit, then express it as a multiple of three standard deviations. If that distance equals exactly three standard deviations the index is 1.0.

μ Mean of the simulated response.

σ Standard deviation of the simulated response.

USL, LSL Upper and lower specification limits. For film thickness both apply, at 8 and 12 μm . For ΔE_{00} only an upper limit applies, at 1.0, because there is no such thing as too little colour error. The second form of Equation (17) is used in that case.

3σ The conventional denominator. A normal distribution places 99.73% of its values within $\pm 3\sigma$ of the mean.

3.5.1 Ranking the inputs

Process inputs are ranked by **standardised regression coefficients**, obtained by regressing the response on inputs that have first been converted to standard scores:

$$\beta_i = \frac{\partial z_y}{\partial z_{x_i}}, \quad z_q = \frac{q - \bar{q}}{\sigma_q} \quad (18)$$

Why this ranking and not another. A raw sensitivity would report how much the colour changes per degree of temperature. That is incomplete, because it ignores how much the temperature actually varies in practice. The standardised coefficient reports how much the colour changes per *standard deviation* of temperature, and therefore folds together the physical sensitivity and the current control spread. That combination is precisely what determines which tolerance is worth money to tighten.

β_i Standardised regression coefficient for input i . Dimensionless.

z_q Standard score of quantity q , obtained by subtracting its mean and dividing by its standard deviation.

$\bar{q}, \sigma_q$ Mean and standard deviation of q .

3.6 Experimental design

A Box-Behnken design is used throughout. For k factors it takes every pair of factors to all four combinations of their high and low levels while holding all remaining factors at their centre value, plus replicated centre points.

Why Box-Behnken rather than a central composite design. A central composite design requires axial points that lie *outside* the stated factor range. For a real process bath this is often impossible: the design would demand acid concentrations or temperatures the line cannot physically hold, or that would damage the parts. A Box-Behnken design places no run outside the stated range, at the cost of estimating the extreme

corners less precisely.

The run count is $2k(k - 1)$ factor points plus centre replicates, giving 15 runs for three factors and 27 for four. A full quadratic response surface is then fitted:

$$y = b_0 + \sum_i b_i x_i + \sum_i b_{ii} x_i^2 + \sum_{i < j} b_{ij} x_i x_j \quad (19)$$

- y Predicted response, either ΔE_{00} or film thickness.
- b_0 Intercept, the predicted response at the centre of the design.
- b_i Linear coefficients. These capture the main effect of each factor.
- b_{ii} Quadratic coefficients. These capture curvature, meaning cases where the response has a minimum or maximum inside the factor range rather than simply rising or falling across it.
- b_{ij} Interaction coefficients. These capture cases where the effect of factor i depends on the level of factor j .
- x_i Factor levels in **coded** units, scaled so that the low level is -1 and the high level is $+1$. Coding matters: on coded levels the coefficients are directly comparable, so the largest coefficient genuinely identifies the largest effect. On raw engineering units this would not hold, because a coefficient in degrees Celsius cannot be compared with one in grams per litre.

4 Verification

No result is reported before the implementation has been checked against independently published values. The automated test suite contains 62 checks. The load bearing ones are recorded below.

Check	Result and significance
All 13 CIEDE2000 reference pairs of Sharma, Wu and Dalal (2005)	Reproduced to within 10^{-4} . These pairs were constructed specifically to exercise the hue wraparound and rotation terms that naive implementations of Equation (7) get wrong.
Perfect reflecting diffuser under D65	Returns exactly $L^* = 100$, $a^* = b^* = 0$, confirming the normalisation of Equation (2) is correct.
D65 reference white point	(95.047, 100, 108.883), matching the CIE tabulation exactly.
$0 \leq R \leq 1$ for every stack at every angle	Passes. This is the check that detected the sign convention error described in Section 3.4.1.
Nominal Type II recipe: 1.5 A/dm ² , 30 min, 20°C, 180 g/L	Returns 9.66 μm against the industry figure of approximately 10 μm . This agreement is the calibration anchor for the entire anodizing model.
Temperature dependence of back dissolution	Verified to fall between $1.8\times$ and $3.0\times$ per 10°C, consistent with the accepted rule of thumb of approximately doubling.
Response surface fitter applied to a known quadratic	Recovers every coefficient of Equation (19) exactly, with $R^2 = 1$.
Opaque film on two different substrates	Returns the same colour to within $\Delta E_{00} = 0.1$, confirming the optical opacity argument used in Section 3.4.3.

Table 1: Verification summary. All checks are automated and execute in continuous integration on Python 3.10 to 3.12.

5 Results

5.1 Anodizing route

At the nominal process window with a black dye, the process is comfortably capable in colour and marginal in thickness.

The Box-Behnken response surface fits thickness essentially perfectly, with $R^2 = 0.99999$, and fits colour well, with $R^2 = 0.932$. The thickness model is dominated by linear terms in current density, bath temperature and acid concentration, exactly as Equation (8) requires. This agreement between an independently fitted empirical surface and the governing equation is itself a useful consistency check.

Response	Mean	σ	Specification	C_{pk}
ΔE_{00} against colour master	0.032	0.019	≤ 1.0	17.04
Oxide thickness	9.653 μm	0.307 μm	8 to 12 μm	1.80
Combined yield				100.00 %

Table 2: Anodizing capability from 6000 Monte Carlo samples.

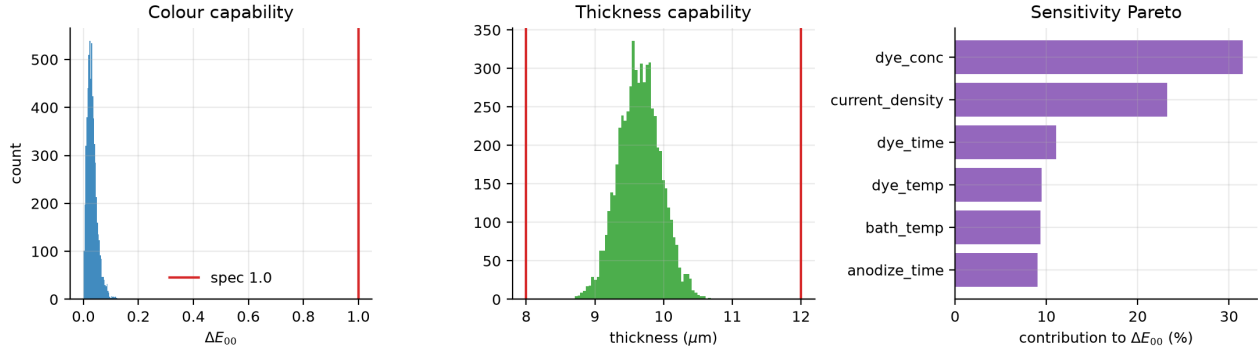


Figure 3: Anodizing capability. Colour (left) sits far inside its specification limit. Thickness (centre) consumes most of its available window. The Pareto chart (right) identifies dye concentration and current density as the leading contributors to colour variation.

5.2 The effect of colour selection

The most consequential result in this study arises from changing nothing at all about the process. Holding the process window, the control tolerances and the sample size fixed, and varying only which dye is in the tank, produces the following:

Dye	Mean ΔE_{00}	σ	C_{pk}	Yield
Black	0.032	0.019	16.68	100.00 %
Graphite grey	0.203	0.158	1.68	99.96 %
Gold or bronze	0.549	0.415	0.36	85.20 %
Red	0.566	0.427	0.34	84.24 %
Blue	0.668	0.512	0.22	77.72 %

Table 3: Identical process, identical tolerances, different dye.

The explanation lies in Equation (10). A saturated black sits deep in Beer-Lambert saturation: the exponential has already fallen close to zero, so further variation in dye loading moves the reflectance hardly at all. Pale and chromatic finishes sit on the steep portion of the same exponential, where identical variation produces a visible shift.

Implication. Yield swings by 22 percentage points with no process change whatsoever. The decision responsible was taken in a design review, by someone selecting a colour, long before any process engineer saw the part. This argues for capability analysis being performed *at the point of colour selection*, not after the line has been commissioned.

5.3 PVD route

The PVD interference stack fails at tolerances that would be entirely normal on a production tool.

A negative capability index means the process mean already lies outside the specification limit. The average part is a reject, which is a materially different situation from a process that is merely too variable.

The response surface for ΔE_{00} fits well, with $R^2 = 0.996$ and adjusted $R^2 = 0.989$, and is dominated by *quadratic* terms in stoichiometry. Physically this means the colour penalty grows faster than linearly as nitrogen control drifts, and it does so in both directions. Drifting nitrogen rich is penalised as heavily as drifting nitrogen lean.

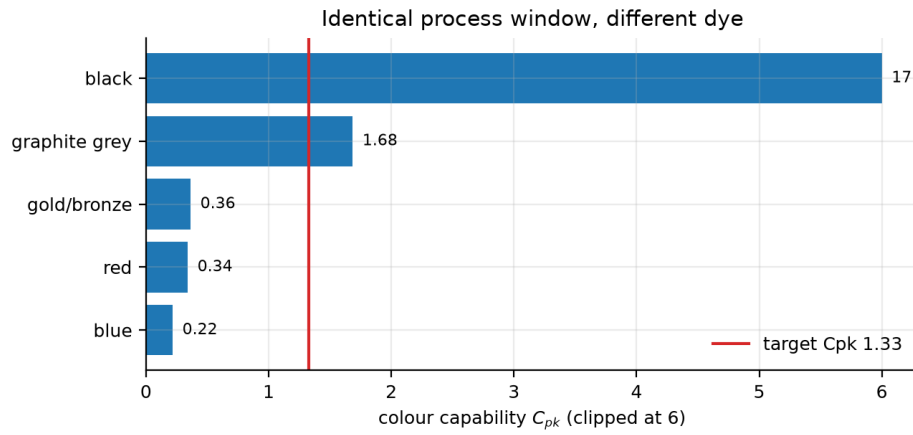


Figure 4: Colour capability by dye choice under an identical process window. Only black and graphite clear the $C_{pk} = 1.33$ gate.

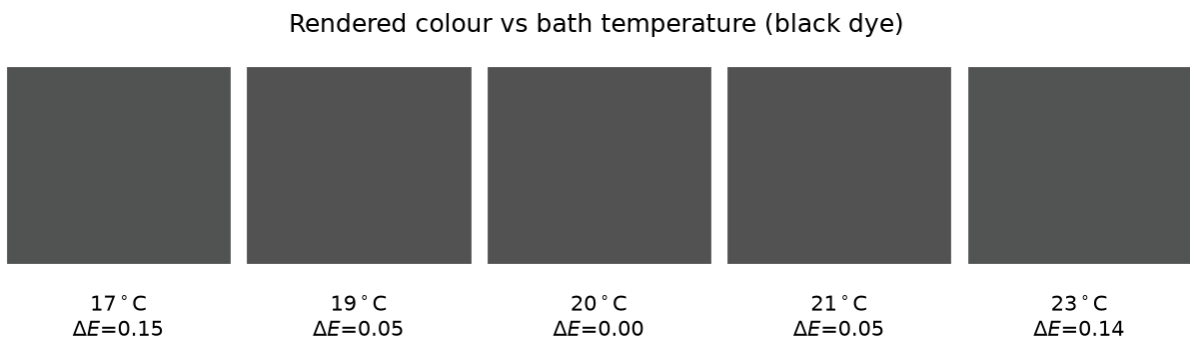


Figure 5: Rendered appearance across a $\pm 3^\circ\text{C}$ bath excursion, with the corresponding ΔE_{00} . Swatches are computed from the simulated spectra through the same colourimetric pipeline, then converted to sRGB for display.

Quantity	Value	Variance contribution	
Mean ΔE_{00}	1.042	Stoichiometry	69.8 %
$\sigma_{\Delta E_{00}}$	0.773	Overcoat thickness	27.8 %
Colour C_{pk}	-0.018	TiN thickness	2.5 %
Yield against $\Delta E_{00} \leq 1.0$	56.4 %		

Table 4: PVD capability for the stack air / SiO_2 90 nm / TiN 300 nm / Ti 80 nm / SS304, from 6000 samples.

Applying the greedy tightening procedure, which repeatedly identifies the dominant contributor and reduces its tolerance, converges after 18 iterations:

Control	Current	Required	Change
TiN stoichiometry, set by nitrogen flow	± 0.040	± 0.0067	6.0 $\times$ tighter
SiO_2 overcoat thickness	± 8.0 nm	± 3.0 nm	2.7 $\times$ tighter
TiN thickness	± 20 nm	± 20 nm	unchanged

Table 5: Control limits required to achieve $C_{pk} = 1.33$.

The counter-intuitive result. TiN thickness is the variable an engineer would instinctively control, and it is almost irrelevant. It accounts for 2.5 % of the colour variance and its tolerance never requires tightening. The film is optically opaque beyond roughly 150 nm, so depositing more of it changes nothing an observer can detect. Composition is the real lever, and composition is governed by a gas flow controller, not by a deposition timer.

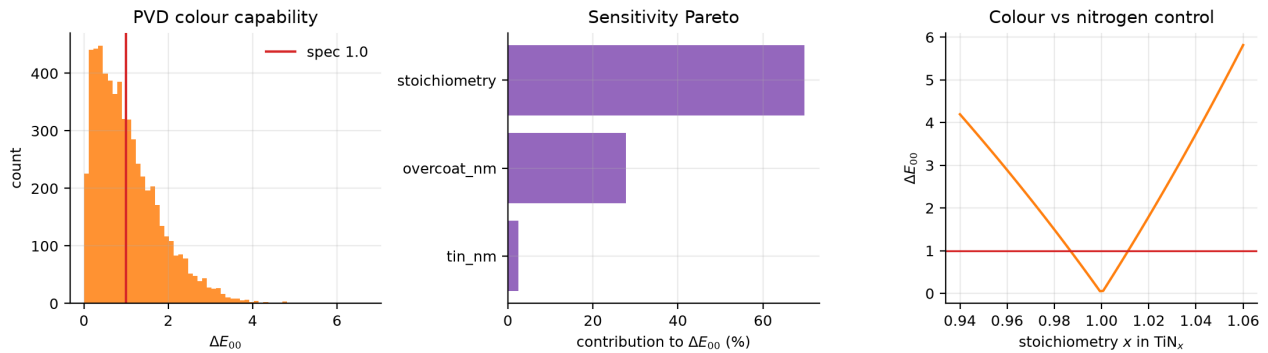


Figure 6: PVD capability. A substantial fraction of the simulated distribution (left) lies beyond the specification limit. Stoichiometry dominates the Pareto chart (centre). The colour penalty against nitrogen control (right) is steeply quadratic, so symmetric drift in either direction is penalised.

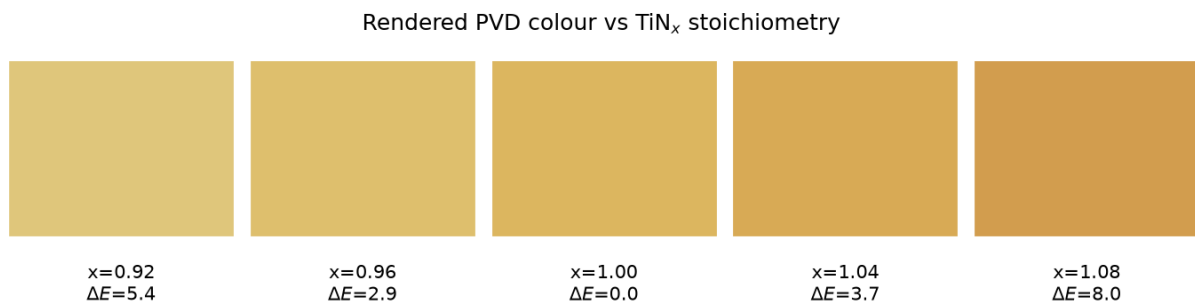


Figure 7: Rendered PVD appearance across the stoichiometry range. An 8 % composition swing carries the finish from pale champagne to warm bronze.

5.4 Comparison of the two routes

	Anodize with black dye	PVD interference stack
Binding constraint	Film thickness	Composition
Colour C_{pk}	16.68	−0.018
Limiting C_{pk}	1.80, from thickness	−0.018, from colour
Dominant variable	Current density and dye concentration	Nitrogen stoichiometry
Angular colour shift	Negligible, being a bulk absorber	Significant, being an interference stack
What to instrument	Bath temperature control, coating thickness gauge	Mass flow control, in chamber optical monitor

Table 6: Comparative failure modes of the two finishing routes.

6 Discussion

6.1 Why a control plan does not transfer between routes

The comparison in Table 6 shows that the two routes fail in entirely different places. The anodizing line requires thickness metrology and temperature control. The PVD line requires gas flow precision and in chamber optical monitoring. Neither conclusion is reachable without modelling the underlying physics, because both routes produce what a process sheet would describe as a coloured part, and their documentation looks superficially similar.

This has a practical consequence. An organisation that has developed process control expertise on one route should not assume that expertise transfers. The instruments differ, the critical variables differ, and the relationship between the two differs.

6.2 Why thickness is the wrong instinct for PVD

The result that TiN thickness contributes only 2.5 % of colour variance deserves emphasis, because it contradicts a reasonable engineering intuition. Thickness is easy to measure, easy to control, and in most coating contexts it is the primary specification. Here it is nearly irrelevant.

The reason is optical opacity. Equation (15) gives titanium nitride an extinction coefficient of order 2 across the visible range, which corresponds to an absorption depth of tens of nanometres. A 300 nm film is therefore many absorption depths thick. Light that enters it does not return. Adding a further 50 nm changes nothing an observer can detect, because that additional material sits behind a region light never reaches.

6.3 Why the colour selection result matters most

Of the findings in Section 5, the dye comparison has the largest practical consequence, because it identifies a lever that costs nothing to use and is available earliest.

Every other finding in this report describes an improvement that must be bought: tighter mass flow controllers, better temperature control, additional metrology. The colour selection finding describes an improvement that is obtained by making a different decision at no capital cost, provided it is made before the finish is committed. A 22 percentage point yield difference is larger than most process improvement programmes deliver.

The organisational obstacle is that the decision sits with industrial design, while the consequence lands on manufacturing. Making that connection visible early is a process engineering responsibility even though the decision is not a process engineering decision.

7 Limitations

These results are derived entirely from simulation. Stating the boundaries precisely is part of the result, and each limitation below is stated with its consequence for how the findings may be used.

7.1 Optical constants are representative, not measured

The Drude-Lorentz parameters in Equation (16) are representative literature fit values for each material class. They are not measurements of any specific supplier's coating.

Consequence. They reproduce the correct colour family and the correct dispersion shape, which is what a *sensitivity* study requires. The conclusions of this report concern how strongly colour responds to process variation, and that behaviour is governed by stack geometry and by the shape of $n(\lambda)$, not by the third decimal place of any oscillator parameter. The absolute $L^*a^*b^*$ values reported here should *not* be treated as predictions of a physical coating. Absolute colour prediction against a real colour master would require measured optical constants, which the software accepts as a drop in replacement.

7.2 Empirical relations in the anodizing model

The porosity and dye uptake relations, Equation (9) in particular, are phenomenological. They were constructed to reproduce published Type II behaviour rather than derived from first principles.

Consequence. Trends are trustworthy. Absolute dye loadings are not.

7.3 Capability computed here is short term

Samples are drawn at fixed nominal settings, so what Equation (17) computes is within batch, short term capability.

Consequence. Long term capability on a real line will be worse, because it additionally carries drift between bath changes, variation with rack position, differences between operators and shifts, and gradual bath ageing. The figures in Section 5 should therefore be read as an optimistic bound. A process that achieves $C_{pk} = 1.33$ in this model would require further margin to achieve it in production.

7.4 No experimental validation

No coupons were anodized, no stacks were deposited, and no spectrophotometer readings were taken.

Consequence. This is the single largest gap in the study and the natural next step. A validation exercise of nine anodized coupons, spanning three bath temperatures and three current densities with colour measured on a calibrated instrument, would be sufficient to confirm or correct the anodizing half. The software is deliberately structured so that a measured dataset drops directly into the existing comparison path.

8 Conclusions and Recommendations

8.1 Conclusions

1. A single pipeline running from physics to colour to capability can serve two physically unrelated finishing routes. This makes them directly comparable on the metric that matters commercially, which is yield against a colour specification.
2. The PVD interference stack examined here is not capable at realistic tolerances, yielding 56.4 % with $C_{pk} = -0.018$. Achieving $C_{pk} = 1.33$ requires nitrogen control six times tighter, and requires no improvement whatsoever in the thickness control an engineer would instinctively tighten first.
3. Anodized black is robust in colour and limited by film thickness. The constraint, and therefore the entire control strategy, is the opposite of the PVD case.
4. Cosmetic capability should be assessed at the point of colour selection. Changing only the dye, with the process held entirely fixed, moved yield from 100 % to 77.7 %. That effect is larger than most process improvements could recover, and it is free to avoid provided it is considered early enough.

8.2 Recommendations

1. **Introduce a capability screen at colour selection.** Before a finish is committed, evaluate the predicted capability of the proposed colour under the intended process window. Reserve pale and chromatic finishes for applications where the tighter process control is justified.
2. **For PVD, prioritise mass flow control.** Investment should be directed towards nitrogen partial pressure precision and towards in chamber optical monitoring. Investment in deposition timing precision will not improve colour yield.
3. **For anodizing, prioritise bath temperature control and thickness metrology.** These address the two mechanisms that Equation (8) identifies as binding.
4. **Validate experimentally before relying on absolute values.** A nine coupon anodizing trial with calibrated colour measurement is sufficient to confirm the anodizing model. Until that is done, use the model for ranking and for sensitivity, not for absolute colour prediction.
5. **Extend the model to long term capability** once production data is available, so that between batch drift can be included and the optimistic bound of Section 7.3 removed.

A Nomenclature

Symbol	Meaning	Units
<i>Colourimetry</i>		
λ	Wavelength of light	nm
$R(\lambda)$	Reflectance factor	dimensionless, 0 to 1
$S(\lambda)$	Illuminant spectral power distribution	relative
$\bar{x}, \bar{y}, \bar{z}$	CIE 1931 colour matching functions	dimensionless
X, Y, Z	Tristimulus values	dimensionless
X_n, Y_n, Z_n	Reference white tristimulus values	dimensionless
k	Colourimetric normalising constant	dimensionless
L^*	Lightness	0 to 100
a^*	Green to red coordinate	dimensionless
b^*	Blue to yellow coordinate	dimensionless
ΔE_{00}	CIEDE2000 colour difference	dimensionless
S_L, S_C, S_H	CIEDE2000 weighting functions	dimensionless
R_T	CIEDE2000 rotation term	dimensionless
k_L, k_C, k_H	CIEDE2000 parametric factors	dimensionless
<i>Anodizing</i>		

Symbol	Meaning	Units
x	Oxide film thickness	m, reported in μm
t	Time	s, reported in min
M_{ox}	Molar mass of Al_2O_3	0.10196 kg/mol
j	Current density	A/m^2 , reported in A/dm^2
η	Current efficiency	dimensionless, 0 to 1
z	Electrons per formula unit	6
F	Faraday constant	96 485 C/mol
ρ_{ox}	Anodic oxide density	3100 kg/m^3
k_0	Dissolution pre-exponential factor	m/s per g/L
E_a	Dissolution activation energy	65 000 J/mol
R_g	Universal gas constant	8.314 J/(mol K)
T	Absolute temperature	K, reported in $^\circ\text{C}$
$[\text{H}_2\text{SO}_4]$	Sulfuric acid concentration	g/L
m	Reaction order in acid	1
P	Film porosity	volume fraction
V	Cell voltage	V
$\varepsilon(\lambda)$	Dye molar absorptivity	relative
c	Dye concentration in film	relative
t_d	Dye immersion time	min
τ	Dye uptake time constant	min
R_{surf}	Air to oxide Fresnel reflectance	dimensionless
R_{Al}	Aluminium substrate reflectance	dimensionless
<i>Thin film optics</i>		
N	Complex refractive index, $n - ik$	dimensionless
n	Real refractive index	dimensionless
k	Extinction coefficient	dimensionless
M_j	Characteristic matrix of layer j	dimensionless
δ_j	Phase thickness of layer j	radians
η_j	Optical admittance of layer j	dimensionless
θ_j	Propagation angle in layer j	radians, complex
t_j	Physical thickness of layer j	nm
B, C	Normalised field amplitudes	dimensionless, complex
Y	Equivalent surface admittance	dimensionless, complex
ϵ	Complex dielectric function	dimensionless
ϵ_∞	High frequency dielectric constant	dimensionless
ω	Angular frequency of light	eV
ω_p	Plasma frequency	eV
Γ	Drude damping constant	eV
f_j	Lorentz oscillator strength	dimensionless
ω_j	Lorentz resonant frequency	eV
γ_j	Lorentz damping	eV
x	Stoichiometry in TiN_x	dimensionless
<i>Statistics</i>		
μ	Mean of a response	response units
σ	Standard deviation	response units
USL, LSL	Upper and lower specification limits	response units

Symbol	Meaning	Units
C_{pk}	Process capability index	dimensionless
β_i	Standardised regression coefficient	dimensionless
z_q	Standard score of quantity q	dimensionless
b_0, b_i, b_{ii}, b_{ij}	Response surface coefficients	response units
R^2	Coefficient of determination	0 to 1

B Reproducing this study

Every number and every figure in this report is regenerated by a single command:

```
python scripts/run_study.py
```

The test suite described in Section 4 is executed by `python -m pytest` and comprises 62 checks. Both the tests and the full study are re-executed in continuous integration on every change, across Python 3.10, 3.11 and 3.12, so this document and the code that produced it cannot drift apart.

The design tables are additionally exported in a form suitable for JMP, together with a JSL script that reconstructs the response surface analysis, the prediction profiler and the contour profiler directly.

Output	Location
Numerical results, all values	outputs/results.json
Anodizing design table	outputs/anodize_doe.csv
PVD design table	outputs/pvd_doe.csv
Monte Carlo samples	outputs/*_montecarlo.csv
Dye comparison	outputs/anodize_dye_robustness.csv
JMP scripts	outputs/*.jsl
Figures used in this report	docs/figures/

Table 8: Where each output is written.

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